OPMIG-04: OpenPipeline Migration Guide: Part 4

> **Series:** OPMIG — OpenPipeline Migration | **Notebook:** 4 of 9 | **Created:** December 2025 | **Last Updated:** 01/28/2026

Pipeline Configuration Fundamentals

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Learning Objectives

By the end of this notebook, you will:

- ✅ Navigate the OpenPipeline UI confidently

- ✅ **Follow step-by-step UI walkthrough**
- ✅ Create custom pipelines from scratch
- ✅ Configure processors with best practices
- ✅ Test configurations with sample data
- ✅ **Build complete end-to-end pipeline examples**
- ✅ **Configure pipelines via API/code**
- ✅ Verify pipeline processing with DQL

Accessing OpenPipeline Settings - Detailed Guide

Navigation Path

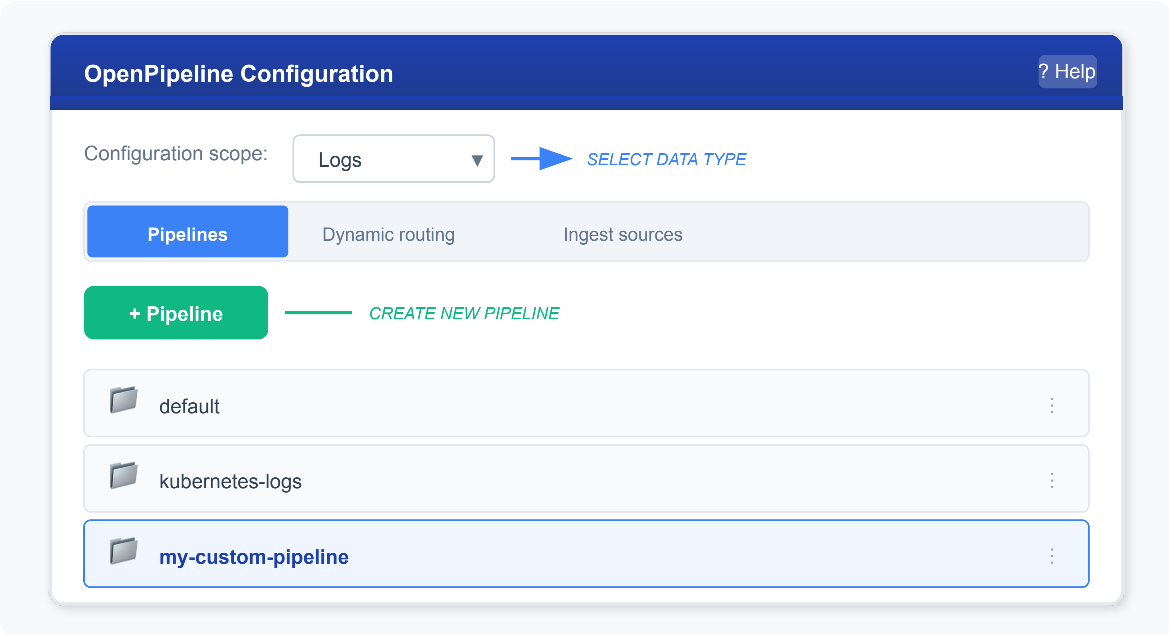
From Dynatrace Home:

```
1. Click ≡ (hamburger menu) in top-left
2. Scroll to "Manage" section
3. Click "Settings"
4. In left sidebar, expand "Process and contextualize"
5. Click "OpenPipeline"
```

Direct URL:

```
https://{your-environment}.apps.dynatrace.com/ui/settings/openpipeline
```

OpenPipeline UI Overview



Configuration Scopes

The dropdown at the top lets you switch between data types:

Scope	Icon	What It Handles
Logs	📋	Log records from all sources
Spans	🔗	Distributed trace spans
Metrics	📊	Metric data ingestion
Events	⚡	Platform events
Business Events	💼	Business analytics events
Security Events	🔐	Security data

> 💡 **Tip:** Each scope has its own separate set of pipelines. Start with **Logs** for your migration.

Three Main Tabs

1. Pipelines Tab

- View all pipelines for selected scope
- Create new pipelines
- Edit pipeline configuration
- Configure processing, extraction, storage

2. Dynamic Routing Tab

- Create routing rules
- Map data to pipelines
- Set matching conditions
- Control route order

3. Ingest Sources Tab

- View data sources
- See ingestion statistics
- Monitor source health
- Troubleshoot ingestion issues

Step-by-Step: Creating Your First Pipeline

Let's walk through creating a complete pipeline for nginx access logs.

Step 1: Create the Pipeline

Actions:
1. Go to OpenPipeline → Logs scope
2. Click **[+ Pipeline]** button (top-left)
3. Enter name: `nginx-access-logs`
4. Add description: `Process nginx access logs with parsing and metric extraction`
5. Click **Create**

Result: New empty pipeline appears in list

Step 2: Configure Processing Stage

2a. Add DQL Processor for Parsing

Actions:
1. Click on your `nginx-access-logs` pipeline
2. Go to **Processing** tab
3. Click **[+ Processor]** button
4. Select **DQL** from processor types

Configuration:

```
Name: Parse nginx access log

Matching condition:
log.source == "nginx" OR contains(content, "GET") OR contains(content,
"POST")

Processor definition:
parse content, "IPADDR:client_ip SPACE '-' SPACE LD:user SPACE '['
LD:timestamp ']' SPACE '\"' LD:method SPACE LD:path SPACE LD:protocol '\"'
SPACE INT:status_code SPACE INT:bytes"
```

Test with Sample Data:

```
{
  "content": "192.168.1.100 - - [12/Dec/2024:10:30:45 +0000] \"GET /api/users
HTTP/1.1\" 200 1234",
  "log.source": "nginx"
}
```

5. Click **[Run sample data]**
6. Verify fields extracted: `client_ip`, `method`, `path`, `status_code`, `bytes`
7. Click **[Save]**

2b. Add Enrichment Processor

Actions:
1. Click **[+ Processor]** again

2. Select **DQL**

Configuration:

```
Name: Add environment tags

Matching condition:
(leave empty to apply to all)

Processor definition:
fieldsAdd environment = "production"
| fieldsAdd service_type = "web-server"
| fieldsAdd response_category = if(status_code &lt; 400, "success",
                                else: if(status_code &lt; 500,
"client_error",
                                else: "server_error"))
```

3. Click **[Save]**

Step 3: Configure Metric Extraction

Actions:
1. Go to **Metric extraction** tab
2. Click **[+ Metric]**

Configuration for Request Count:

```
Metric type: Counter
Metric key: log.nginx.request_count

Dimensions:
  - method
  - status_code
  - environment

Matching condition:
isNotNull(status_code)
```

3. Click **[Save]**
4. Click **[+ Metric]** again for bytes metric

Configuration for Response Size:

```
Metric type: Value
Metric key: log.nginx.response_bytes
Value field: bytes

Dimensions:
  - method
  - path

Matching condition:
isNotNull(bytes)
```

5. Click **[Save]**

Step 4: Configure Storage

Actions:
1. Go to **Storage** tab
2. Select target bucket: `default_logs`
3. Click **[Save]**

> 💡 **Tip:** If you created custom buckets, select the appropriate one here (e.g., `web_logs` with 14-day retention)

Step 5: Create Dynamic Routing

Actions:
1. Go to **Dynamic routing** tab (top of OpenPipeline page)
2. Click **[+ Dynamic route]**

Configuration:

```
Name: Route nginx logs

Matching condition:
log.source == "nginx" OR contains(content, "nginx")

Pipeline: nginx-access-logs
```

3. Click **[Save]**

Step 6: Test End-to-End

Option 1: Send Test Log via API

```
curl -i -X POST \
  "https://{your-environment}.live.dynatrace.com/api/v2/logs/ingest" \
  -H "Content-Type: application/json" \
  -H "Authorization: Api-Token {your-token}" \
  -d '{
    "content": "192.168.1.100 - - [12/Dec/2024:10:30:45 +0000] \"GET
/api/users HTTP/1.1\" 200 1234",
    "log.source": "nginx"
  }'
```

Option 2: Check with Existing Logs

Wait 2-3 minutes for processing, then run verification query (see next section).

Step 7: Verify Processing

See the validation queries in the next section to confirm:
- Logs are routed to your pipeline ✓
- Fields are parsed correctly ✓
- Metrics are being generated ✓

Complete Pipeline Examples

Example 1: Application Logs with Error Tracking

Use Case: Java application logs with structured format

Pipeline: `java-app-logs`

Processing Stage:

Processor 1: Parse Log Structure

```
parse content, "'[' TIMESTAMP('yyyy-MM-dd HH:mm:ss'):log_time ']' SPACE '['
LD:level ']' SPACE '[' LD:thread ']' SPACE LD:class ' - ' DATA:message"
```

Processor 2: Extract Request ID

```
parse content, "'requestId=' LD:request_id"
```

Processor 3: Classify Errors

```
fieldsAdd error_type = if(contains(message, "NullPointerException"), "NPE",
                       else: if(contains(message, "SQLException"), "DB",
                       else: if(contains(message, "TimeoutException"),
"Timeout",
                       else: "Other")))
| fieldsAdd severity = if(level == "ERROR", "critical",
                       else: if(level == "WARN", "warning",
                       else: "info"))
```

Metric Extraction:
- **Counter:** `log.java.error_count` (dimensions: error_type, class)
- **Counter:** `log.java.request_count` (dimensions: level, thread)

Event Extraction:
- **Event Name:** `application.error`
- **Description:** `Error in {class}: {error_type}`
- **Type:** ERROR
- **Matching:** `level == "ERROR"`

Storage: `default_logs` (35 days)

Example 2: Payment Service with PCI-DSS Compliance

Use Case: Financial logs requiring credit card masking

Pipeline: `payment-service-secure`

Processing Stage (in order):

Processor 1: Mask Credit Cards (FIRST!)

```
fieldsAdd content = replacePattern(content, "CREDITCARD", replacement: "
[CC_REDACTED]")
```

Processor 2: Mask CVV

```
fieldsAdd content = replacePattern(content, "('cvv='|'cvc=') INT{3,4}",
replacement: "cvv=[REDACTED]")
```

Processor 3: Parse Payment Data

```
parse content, "'orderId=' INT:order_id ','"
| parse content, "'amount=' DOUBLE:amount"
| parse content, "'currency=' LD:currency"
| parse content, "'status=' LD:payment_status"
```

Processor 4: Add Tags

```
fieldsAdd service = "payment"
| fieldsAdd compliance = "pci-dss"
| fieldsAdd audit_required = true
```

Metric Extraction:
- **Value:** `log.payment.amount` (value: amount, dimensions: currency, payment_status)
- **Counter:** `log.payment.transaction_count` (dimensions: payment_status, currency)

Business Event Extraction:
- **Type:** `com.company.payment.processed`
- **Provider:** `payment-service`
- **Fields:** order_id, amount, currency, payment_status

Storage: `financial_logs` (90 days - compliance requirement)

Example 3: Kubernetes Logs with Pod Context

Use Case: Container logs with K8s metadata

Pipeline: `kubernetes-app-logs`

Processing Stage:

Processor 1: Drop System Logs

```
Matching: k8s.namespace.name == "kube-system" AND loglevel == "DEBUG"
Action: Drop
```

Processor 2: Parse JSON Logs

```
parse content, "JSON:json_payload"
```

Processor 3: Enrich with K8s Context

```
fieldsAdd app = k8s.deployment.name
| fieldsAdd namespace = k8s.namespace.name
| fieldsAdd pod_name = k8s.pod.name
| fieldsAdd container = k8s.container.name
```

Metric Extraction:
- **Counter:** `log.k8s.pod_log_count` (dimensions: namespace, app, loglevel)

Storage: Route based on namespace:
- production namespace → `prod_logs` (35 days)
- staging namespace → `staging_logs` (14 days)
- dev namespace → `dev_logs` (7 days)

Understanding the Interface

Pipeline Configuration Tabs

Each pipeline has these configuration sections:

Tab	Purpose	Key Actions
Processing	Transform and enrich data	Add DQL processors, parsers
Metric extraction	Create metrics from data	Define value/counter metrics
Event extraction	Generate events	Create platform events
Bizevent extraction	Create business events	Generate bizevents
Storage	Configure bucket routing	Set target bucket

Processor Configuration Panel

When adding a processor, you configure:

1. **Name**: Descriptive name for the processor
2. **Matching condition**: When this processor runs
3. **Processor definition**: What the processor does
4. **Sample data**: Test data to validate

Add processor: DQL

Name:

```
Parse nginx access log
```

Matching condition:

```
log.source == "nginx"
```

Processor definition:

```
parse content, "IPADDR:client_ip SPACE '-' SPACE
LD:user SPACE '[' LD:timestamp ']' ..."
```

Sample data:

```
{"content": "192.168.1.1 - - [10/Dec/2024:10:30:45
+0000] \"GET /api/users HTTP/1.1\" 200 1234"}
```

Run sample data | Cancel | Save

Creating Your First Pipeline

Step-by-Step: Create a Log Processing Pipeline

Example: Create a pipeline for nginx access logs

Step 1: Navigate to OpenPipeline

```
Settings → Process and contextualize → OpenPipeline → Logs
```

Step 2: Create Pipeline

1. Click **+ Pipeline**
2. Enter name: `nginx-access-logs`
3. Click **Create**

Step 3: Add Processing (Parsing)

1. Go to **Processing** tab
2. Click **+ Processor** → **DQL**
3. Configure:
   - **Name**: `Parse nginx access log`
   - **Matching condition**: `matchesValue(content, "*HTTP*")`
   - **Processor definition**:
   `parse content, "IPADDR:client_ip SPACE '-' SPACE LD:user SPACE '[' LD:timestamp ']' SPACE '\"' LD:method SPACE LD:path SPACE LD:protocol '\"' SPACE INT:status_code SPACE INT:bytes"`

Step 4: Test with Sample Data

1. Enter sample log:
   `json {"content": "192.168.1.100 - - [12/Dec/2024:10:30:45 +0000] \"GET /api/users HTTP/1.1\" 200 1234"}`
2. Click **Run sample data**
3. Verify extracted fields

Step 5: Save Pipeline

Click **Save** to store the configuration.

Configuring Processors

DQL Processor Examples

The DQL processor is the most flexible. Here are common patterns:

Add Static Fields

```
fieldsAdd environment = "production"
fieldsAdd application = "web-frontend"
```

Add Conditional Fields

```
fieldsAdd severity = if(loglevel == "ERROR", "critical",
                     else: if(loglevel == "WARN", "warning",
                     else: "normal"))
```

Parse with DPL Pattern

```
parse content, "'userId=' LD:user_id ','"
```

Remove Sensitive Fields

```
fieldsRemove password, secret_key, api_token
```

Rename Fields for Standardization

```
fieldsRename old_field_name = standardized_name
```

Drop Processor Examples

Drop processors remove records matching conditions:

Matching Condition	Effect
`loglevel == "DEBUG"`	Drop all debug logs
`contains(content, "healthz")`	Drop health check logs
`contains(content, "/metrics")`	Drop Prometheus scrapes
`status == "TRACE"`	Drop trace-level logs

Technology Parsers

Built-in parsers for common formats:

Parser	Applies To	Extracted Fields
Apache	Apache access logs	client_ip, method, path, status
Nginx	Nginx access logs	Similar to Apache
JSON	JSON-formatted logs	All JSON fields flattened
Syslog	Syslog format	facility, severity, hostname

Setting Up Dynamic Routing

Dynamic routing sends data to specific pipelines based on matching conditions.

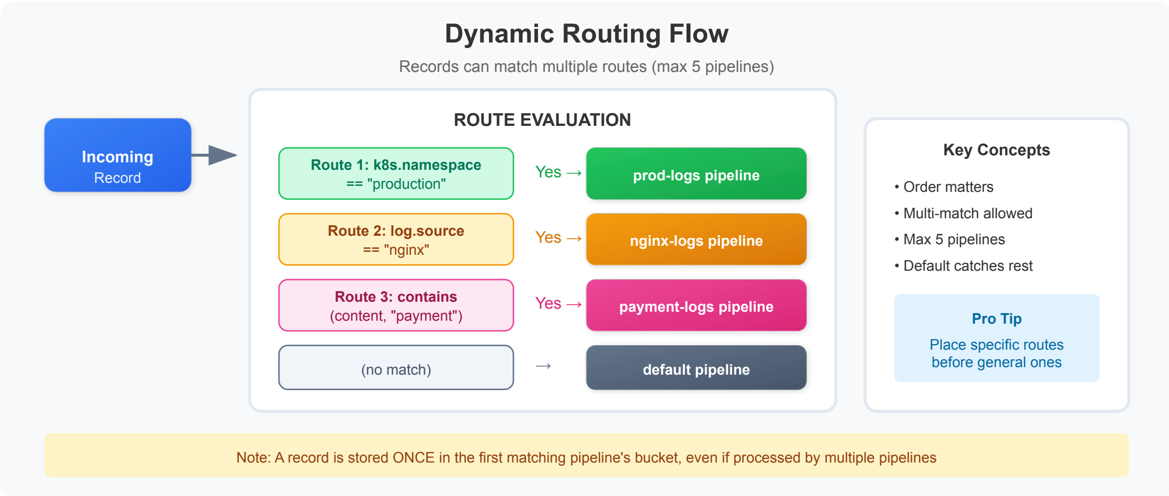
Routing Configuration Steps

1. Go to **Dynamic routing** tab
2. Click **+ Dynamic route**
3. Configure:
   - **Name**: Descriptive route name
   - **Matching condition**: When to use this route
   - **Pipeline**: Target pipeline

Matching Condition Examples

Condition	Routes To
`log.source == "nginx"`	nginx-logs pipeline
`k8s.namespace.name == "production"`	prod-logs pipeline
`contains(content, "payment")`	payment-logs pipeline
`dt.openpipeline.source == "generic"`	api-ingested pipeline
`host.name == "web-server-01"`	specific host pipeline

Route Evaluation Order



> 💡 **Tip:** Routes are evaluated in order. Place more specific routes first, general routes last.

Testing Your Configuration

Using Sample Data in UI

Every processor supports sample data testing:

1. Enter JSON sample in **Sample data** field
2. Click **Run sample data**
3. Review output in preview panel
4. Adjust processor definition as needed

Sample Data Format

```
{
  "content": "Your log message here",
  "log.source": "your-source",
  "timestamp": "2024-12-12T10:30:00Z",
  "k8s.namespace.name": "production"
}
```

Testing via API

Send test logs to validate end-to-end:

```
curl -i -X POST \
  "https://{your-environment}.live.dynatrace.com/api/v2/logs/ingest" \
  -H "Content-Type: application/json" \
  -H "Authorization: Api-Token {your-token}" \
  -d '{"content":"Test log message","log.source":"test-source"}'
```

Response `204` indicates successful ingestion.

Configuration Best Practices

Pipeline Design Principles

Principle	Recommendation
Single Purpose	One pipeline per use case (not one mega-pipeline)
Specific Routing	Use precise matching conditions
Security First	Add masking processors early in the chain
Drop Early	Remove unwanted data before processing
Test Thoroughly	Validate every processor with sample data

Processor Ordering

Within a pipeline, order processors logically:

```
1. Masking      → Protect sensitive data first
2. Drop         → Remove unwanted records
3. Parse        → Extract structured fields
4. Enrich       → Add context fields
5. Transform    → Compute derived values
```

Naming Conventions

Type	Convention	Example
Pipeline	`{source}-{purpose}`	`nginx-access-logs`
Processor	`{action}-{target}`	`parse-request-line`
Route	`route-to-{pipeline}`	`route-to-nginx-logs`

Common Mistakes to Avoid

Mistake	Problem	Solution
Overly broad routing	Too many logs hit pipeline	Use specific conditions
No masking	Sensitive data stored	Add masking before processing
Complex conditions	Hard to maintain	Break into multiple pipelines
Untested patterns	Parsing fails on real data	Always test with samples
Missing fallback	Unknown data lost	Configure default pipeline

---

Verifying Pipeline Processing

After configuring pipelines, verify they're working correctly with these queries.

```
// Verify logs are being processed by your new pipeline
// Replace 'your-pipeline-name' with your actual pipeline name
fetch logs, from: now() - 1h
| filter contains(toString(dt.openpipeline.pipelines), "your-pipeline-name")
| summarize {processed_count = count()}
| fieldsAdd status = if(processed_count &gt; 0, "✅ Pipeline is processing logs", else: "⚠️ No logs processed yet")
```

```
// Check routing distribution across all pipelines
fetch logs, from: now() - 1h
| summarize {record_count = count()}, by: {dt.openpipeline.pipelines}
| sort record_count desc
```

```
// Verify parsing is extracting expected fields
// Check for a specific field that should be parsed
fetch logs, from: now() - 1h
| filter isNotNull(dt.openpipeline.pipelines)
| summarize {
    total = count(),
    with_loglevel = countIf(isNotNull(loglevel)),
    with_status = countIf(isNotNull(status))
  }, by: {dt.openpipeline.pipelines}
| fieldsAdd parsing_rate = round((toDouble(with_loglevel) / toDouble(total))
* 100, decimals: 1)
```

```
// Sample recent logs from your pipeline to inspect parsed fields
fetch logs, from: now() - 1h
| filter contains(toString(dt.openpipeline.pipelines), "your-pipeline-name")
| limit 10
```

```
// Check logs going to default pipeline (may need routing)
// High counts here suggest missing routing rules
fetch logs, from: now() - 1h
| filter isNull(dt.openpipeline.pipelines) OR dt.openpipeline.pipelines == "
[]"
| summarize {unrouted = count()}, by: {log.source}
| sort unrouted desc
| limit 20
```

```
// Monitor pipeline processing over time
fetch logs, from: now() - 24h
| filter isNotNull(dt.openpipeline.pipelines)
| makeTimeseries {record_count = count()}, by: {dt.openpipeline.pipelines},
interval: 1h
```

```
// Verify bucket routing is working
fetch logs, from: now() - 1h
| filter isNotNull(dt.openpipeline.pipelines)
| summarize {record_count = count()}, by: {dt.openpipeline.pipelines,
dt.system.bucket}
| sort record_count desc
```

Complete Pipeline Example

Here's a complete example of a production pipeline configuration:

Pipeline: `payment-service-logs`

Routing Condition:

```
k8s.namespace.name == "payments" OR log.source == "payment-service"
```

Processors (in order):

1. **Mask Credit Cards** (Masking)
   - Matching: `contains(content, "card")`
   - Definition: `fieldsAdd content = replacePattern(toString(content), "\\d{4}[- ]?\\d{4}[- ]?\\d{4}[- ]?\\d{4}", "****-****-****-****")`

2. **Drop Health Checks** (Drop)
   - Matching: `contains(content, "/health") OR contains(content, "/ready")`

3. **Parse Payment Logs** (DQL)
   - Matching: `matchesValue(content, "*transaction*")`
   - Definition: `parse content, "'transaction_id=' LD:transaction_id ',' 'amount=' DOUBLE:amount ',' 'status=' LD:payment_status"`

4. **Add Environment Tag** (DQL)
   - Matching: (none - applies to all)
   - Definition: `fieldsAdd service = "payment-service", criticality = "high"`

Metric Extraction:
- Name: `payment_amount_by_status`
- Field: `amount`
- Dimensions: `payment_status`

Storage:
- Bucket: `financial_logs` (90-day retention)

Next Steps

Now that you can create and configure pipelines, continue with:

Notebook	Focus Area
OPMIG-05	Routing & Bucket Management
OPMIG-06	Processing, Parsing & Transformation
OPMIG-07	Metric & Event Extraction
OPMIG-08	Security, Masking & Compliance

References

- Configure Processing Pipeline
- Processing Examples
- DQL Functions in OpenPipeline
- Dynatrace Pattern Language

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